SMART WASTE COLLECTION AND RECYCLING MANAGEMENT SYSTEM

1. Project Title

Smart Waste Collection and Recycling Management System

The Smart Waste Collection and Recycling Management System is a menu-driven waste management application designed for residential areas. It records waste collection details, classifies waste into plastic, paper, glass, metal and organic categories, calculates recyclable quantities, analyses collection data and generates summary reports. The actual user-facing application is implemented as a standalone Flutter application with local persistence. A Python reference implementation is included to demonstrate the required course outcomes involving strings, functions, mixed data structures, file handling, modules and packages.

2. Pseudocode

```
BEGIN

DEFINE fixed waste categories:
   PLASTIC, PAPER, GLASS, METAL, ORGANIC

LOAD saved waste collection records from local storage

IF no records exist:
   CREATE sample records
   SAVE records

DISPLAY main menu

REPEAT
   DISPLAY:
      1. Add Collection
      2. View Collections
      3. Update Collection
      4. Delete Collection
      5. Search Collection
      6. Classify Waste
      7. Analyse Waste
      8. Generate Report
      9. Exit

   READ user choice

   IF choice = 1:
      READ collection ID, resident, area and date
      READ quantities for plastic, paper, glass, metal and organic
      VALIDATE text and numeric inputs
      CHECK duplicate collection ID
      CALCULATE total waste
```

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    CALCULATE recyclable waste
    CALCULATE recycling percentage
    DETERMINE waste level
    ADD record to collection list
    UPDATE local storage
    DISPLAY success message

ELSE IF choice = 2:
    LOAD/display all records
    DISPLAY total, recyclable and organic quantities

ELSE IF choice = 3:
    READ collection ID
    FIND matching record
    IF found:
        EDIT record fields
        RECALCULATE totals
        UPDATE local storage
    ELSE:
        DISPLAY record not found

ELSE IF choice = 4:
    READ collection ID
    FIND matching record
    IF found:
        CONFIRM deletion
        REMOVE record
        UPDATE local storage
    ELSE:
        DISPLAY record not found

ELSE IF choice = 5:
    READ search keyword
    NORMALIZE keyword using string operations
    SEARCH collection ID, resident, area and category
    DISPLAY matching records

ELSE IF choice = 6:
    DISPLAY waste categories
    READ selected category
    DISPLAY description, examples and disposal recommendation

ELSE IF choice = 7:
    CALCULATE:
        total waste
        recyclable waste
        organic waste
        category totals
        average waste
        recycling percentage
        highest category
        lowest category
        highest waste area
        highest waste resident
    DISPLAY charts/statistics/recommendations

ELSE IF choice = 8:
    GENERATE summary report
    SAVE report locally
    DISPLAY report
```

ELSE IF choice = 9:
SAVE current records
EXIT

ELSE:
DISPLAY invalid choice

UNTIL choice = 9

END

SOURCE CODE:

```
import os

import sys

import datetime


sys.path.insert(0, os.path.dirname(os.path.abspath(__file__)))


import tkinter as tk

from tkinter import ttk, messagebox

import customtkinter as ctk

from matplotlib.backends.backend_tkagg import FigureCanvasTkAgg

import matplotlib

import matplotlib.pyplot as plt

matplotlib.use("TkAgg")


from desktop_app.services.data_manager import DataManager


# ─── Theme
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ctk.set_appearance_mode("Dark")

ctk.set_default_color_theme("green")
```

```
# ——— Color Palette ————————————————————————————————————————————

BG           = "#0a0f1e"      # Deep navy background
SIDEBAR_BG   = "#0d1627"      # Sidebar darker navy
CARD_BG      = "#111c35"      # Card dark blue
CARD_BG2     = "#0f1b30"      # Slightly different card
ACCENT       = "#00d4aa"      # Teal accent
ACCENT2      = "#7c3aed"      # Purple accent
ACCENT3      = "#f59e0b"      # Amber accent
ACCENT4      = "#ef4444"      # Red accent
TEXT_PRI     = "#f0f6ff"      # Primary white text
TEXT_SEC     = "#8aa0c0"      # Muted secondary text
BORDER       = "#1e3a5f"      # Subtle border


CATS    = ["plastic", "paper", "glass", "metal", "organic"]
CAT_CLR = {"plastic": "#3b82f6", "paper": "#f59e0b",
           "glass":   "#10b981", "metal":  "#64748b", "organic": "#a855f7"}


# ——— TTK Table Styling ——————————————————————————————————————————

def apply_table_style():
    style = ttk.Style()
    style.theme_use("clam")
    style.configure("Custom.Treeview",
        background=CARD_BG,
        foreground=TEXT_PRI,
        rowheight=32,
        fieldbackground=CARD_BG,
        bordercolor=BORDER,
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        borderwidth=0,
        font=("Segoe UI", 11)
    )
    style.configure("Custom.Treeview.Heading",
        background="#1e3a5f",
        foreground=ACCENT,
        font=("Segoe UI", 11, "bold"),
        relief="flat",
        borderwidth=0
    )
    style.map("Custom.Treeview",
        background=[("selected", "#1a3a5c")],
        foreground=[("selected", ACCENT)]
    )
    style.configure("Custom.Vertical.TScrollbar",
        background=CARD_BG, troughcolor=SIDEBAR_BG,
        arrowcolor=ACCENT, bordercolor=BORDER
    )


# ═══════════════════════════════════════════════════════════════════════

class SmartWasteApp(ctk.CTk):
    def __init__(self):
        super().__init__()
        self.title("♻  SmartWaste — Smart Waste & Recycling Management System")
        self.geometry("1280x800")
        self.minsize(1100, 720)
        self.configure(fg_color=BG)
```

```
        apply_table_style()


        self.data_manager = DataManager()
        self._active_btn = None

        self._build_layout()
        self.show_dashboard()

    # ── Layout Shell ──────────────────────────────────────────────────────

    def _build_layout(self):
        self.sidebar = tk.Frame(self, bg=SIDEBAR_BG, width=230)
        self.sidebar.pack(side="left", fill="y")
        self.sidebar.pack_propagate(False)

        self.main = tk.Frame(self, bg=BG)
        self.main.pack(side="left", fill="both", expand=True)

        self._build_sidebar()

    def _build_sidebar(self):
        # Logo
        logo_frame = tk.Frame(self.sidebar, bg=SIDEBAR_BG)
        logo_frame.pack(fill="x", pady=(25, 5), padx=18)
        tk.Label(logo_frame, text="♻", font=("Segoe UI", 28), bg=SIDEBAR_BG,
fg=ACCENT).pack(side="left")
        txt_f = tk.Frame(logo_frame, bg=SIDEBAR_BG)
        txt_f.pack(side="left", padx=8)
        tk.Label(txt_f, text="SmartWaste", font=("Segoe UI", 15, "bold"),
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                bg=SIDEBAR_BG, fg=TEXT_PRI).pack(anchor="w")
        tk.Label(txt_f, text="Management System", font=("Segoe UI", 9),
                 bg=SIDEBAR_BG, fg=TEXT_SEC).pack(anchor="w")

        tk.Frame(self.sidebar, bg=BORDER, height=1).pack(fill="x", padx=18, pady=16)

        # Nav buttons
        self._nav_btns = {}
        nav_items = [
            ("dashboard",   "  🏠  Dashboard",          self.show_dashboard),
            ("records",     "  📋  Records",             self.show_records),
            ("add",         "  ➕  Add Collection",      self.show_add_form),
            ("suggestions", "  💡  Smart Suggestions",   self.show_suggestions),
            ("inspector",   "  🔍  Data Structures",     self.show_ds_inspector),
        ]
        for key, label, cmd in nav_items:
            btn = self._nav_btn(self.sidebar, label, cmd, key)
            btn.pack(fill="x", padx=12, pady=3)
            self._nav_btns[key] = btn

        tk.Frame(self.sidebar, bg=BORDER, height=1).pack(fill="x", padx=18, pady=16)

        # Undo Button
        undo_btn = tk.Button(
            self.sidebar, text="  ↩  Undo Last Action",
            bg="#2a1215", fg="#ef4444", activebackground="#3d1a1f",
activeforeground="#ef4444",
            font=("Segoe UI", 12), bd=0, padx=8, pady=10,
            anchor="w", cursor="hand2", relief="flat",
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            command=self.undo_action
        )
        undo_btn.pack(fill="x", padx=12, pady=2)

        # Footer DS Labels
        footer = tk.Frame(self.sidebar, bg=SIDEBAR_BG)
        footer.pack(side="bottom", fill="x", padx=18, pady=20)
        tk.Label(footer, text="Active Data Structures", font=("Segoe UI", 9, "bold"),
                 bg=SIDEBAR_BG, fg=TEXT_SEC).pack(anchor="w")
        for ds in ["• Doubly Linked List", "• Stack (LIFO Undo)", "• Priority Queue
(Heap)",
                   "• Hash Table O(1)", "• Binary Search Tree"]:
            tk.Label(footer, text=ds, font=("Segoe UI", 9),
                     bg=SIDEBAR_BG, fg="#475569").pack(anchor="w")

    def _nav_btn(self, parent, text, cmd, key):
        btn = tk.Button(
            parent, text=text, anchor="w",
            bg=SIDEBAR_BG, fg=TEXT_SEC, activebackground="#152040",
            activeforeground=ACCENT, font=("Segoe UI", 12),
            bd=0, padx=10, pady=10, relief="flat", cursor="hand2",
            command=lambda: self._nav_click(key, cmd)
        )
        return btn

    def _nav_click(self, key, cmd):
        # Deselect old
        if self._active_btn:
            self._nav_btns[self._active_btn].configure(bg=SIDEBAR_BG, fg=TEXT_SEC)
```

```
        # Select new
        self._active_btn = key
        self._nav_btns[key].configure(bg="#152040", fg=ACCENT)
        cmd()

    def clear_main(self):
        for w in self.main.winfo_children():
            w.destroy()

    def _page_header(self, title, subtitle=""):
        hdr = tk.Frame(self.main, bg=BG)
        hdr.pack(fill="x", padx=30, pady=(25, 10))
        tk.Label(hdr, text=title, font=("Segoe UI", 22, "bold"),
                 bg=BG, fg=TEXT_PRI).pack(anchor="w")
        if subtitle:
            tk.Label(hdr, text=subtitle, font=("Segoe UI", 11),
                     bg=BG, fg=TEXT_SEC).pack(anchor="w")
        tk.Frame(self.main, bg=BORDER, height=1).pack(fill="x", padx=30, pady=(0, 5))

    def _card(self, parent, **kwargs):
        return tk.Frame(parent, bg=CARD_BG, **kwargs)

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    # DASHBOARD
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    def show_dashboard(self):
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        self.clear_main()

        records = self.data_manager.get_all_records()

        total_kg = sum(r["total_kg"] for r in records)

        avg_rec  = (sum(r["recyclable_pct"] for r in records) / len(records)) if
records else 0

        unique_areas = len({r["area"] for r in records})


        self._page_header("Dashboard", f"Overview for
{datetime.date.today().strftime('%B %d, %Y')}")


        # ── Metric Cards
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        row = tk.Frame(self.main, bg=BG)

        row.pack(fill="x", padx=30, pady=8)

        metrics = [

            ("📦",  "Total Collected",   f"{total_kg:.1f} kg",       ACCENT),

            ("♻️",  "Avg Recycling Rate",f"{avg_rec:.1f}%",           "#3b82f6"),

            ("📋",  "Total Pickups",     str(len(records)),           ACCENT3),

            ("🗺️",  "Active Zones",      str(unique_areas),          ACCENT2),

        ]

        for icon, label, val, color in metrics:

            card = tk.Frame(row, bg=CARD_BG, bd=0, relief="flat")

            card.pack(side="left", fill="both", expand=True, padx=6, ipady=14,
ipadx=10)

            tk.Label(card, text=icon, font=("Segoe UI Emoji", 20),

                     bg=CARD_BG, fg=color).pack(anchor="w", padx=15, pady=(14, 4))

            tk.Label(card, text=val, font=("Segoe UI", 22, "bold"),

                     bg=CARD_BG, fg=color).pack(anchor="w", padx=15)

            tk.Label(card, text=label, font=("Segoe UI", 10),

                     bg=CARD_BG, fg=TEXT_SEC).pack(anchor="w", padx=15, pady=(2, 14))
```

```
        # ── Charts
        charts_card = tk.Frame(self.main, bg=CARD_BG)
        charts_card.pack(fill="both", expand=True, padx=30, pady=(8, 25))

        cat_totals = {c: 0.0 for c in CATS}
        area_totals = {}
        for r in records:
            for cat in CATS:
                cat_totals[cat] += r.get("waste", {}).get(cat, 0)
            a = r.get("area", "Unknown")
            area_totals[a] = area_totals.get(a, 0) + r["total_kg"]

        plt.rcParams.update({"figure.facecolor": CARD_BG, "axes.facecolor": CARD_BG,
                             "text.color": TEXT_PRI, "axes.labelcolor": TEXT_SEC,
                             "xtick.color": TEXT_SEC, "ytick.color": TEXT_SEC})

        fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(10, 3.8), facecolor=CARD_BG,
                                       gridspec_kw={"wspace": 0.35})

        clrs = [CAT_CLR[c] for c in CATS]
        vals = [cat_totals[c] for c in CATS]
        labels = [c.capitalize() for c in CATS]

        wedges, texts, autotexts = ax1.pie(
            vals, labels=labels, colors=clrs, autopct='%1.0f%%',
            pctdistance=0.78, startangle=140,
            wedgeprops=dict(linewidth=2, edgecolor=CARD_BG)
```

```
        )
        for t in texts: t.set_fontsize(9); t.set_color(TEXT_SEC)
        for a in autotexts: a.set_fontsize(9); a.set_color(TEXT_PRI);
a.set_fontweight("bold")
        ax1.set_title("Category Breakdown", fontsize=12, fontweight="bold",
color=TEXT_PRI, pad=12)

        bars = ax2.bar(list(area_totals.keys()), list(area_totals.values()),
                       color=ACCENT, edgecolor="none", width=0.55)
        for bar in bars:
            h = bar.get_height()
            ax2.text(bar.get_x() + bar.get_width()/2, h + 0.3,
                     f"{h:.0f}kg", ha="center", va="bottom", fontsize=8,
color=TEXT_SEC)
        ax2.set_title("Waste per Zone (kg)", fontsize=12, fontweight="bold",
color=TEXT_PRI, pad=12)
        ax2.tick_params(axis='x', labelsize=8, rotation=20)
        for spine in ax2.spines.values():
            spine.set_color(BORDER)
        ax2.set_facecolor(CARD_BG2)
        ax2.grid(axis="y", color=BORDER, linestyle="--", linewidth=0.5, alpha=0.5)

        canvas = FigureCanvasTkAgg(fig, master=charts_card)
        canvas.draw()
        canvas.get_tk_widget().pack(fill="both", expand=True, padx=5, pady=5)
        plt.close(fig)

    #
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    # RECORDS
```

```
    #
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    def show_records(self):

        self.clear_main()

        self._page_header("Records Manager", "Browse, search and review all waste
collection records")



        # Search Bar

        sb = tk.Frame(self.main, bg=BG)

        sb.pack(fill="x", padx=30, pady=(6, 8))



        self._search_var = tk.StringVar()

        search_bg = tk.Frame(sb, bg=CARD_BG, padx=8, pady=4)

        search_bg.pack(side="left")

        tk.Label(search_bg, text="🔍", bg=CARD_BG, fg=ACCENT, font=("Segoe UI",
12)).pack(side="left")

        entry = tk.Entry(search_bg, textvariable=self._search_var, bg=CARD_BG,

                         fg=TEXT_PRI, insertbackground=ACCENT, relief="flat",

                         font=("Segoe UI", 12), width=35)

        entry.pack(side="left", padx=6, pady=2)

        entry.bind("<Return>", lambda e: self._do_search())



        self._search_btn("Search", self._do_search, ACCENT,
"#0d1627").pack(side="left", padx=8)

        self._search_btn("Clear", self._clear_search, CARD_BG,
TEXT_SEC).pack(side="left")



        # Table

        tbl_card = tk.Frame(self.main, bg=CARD_BG, bd=0)

        tbl_card.pack(fill="both", expand=True, padx=30, pady=(0, 25))
```

```
        cols =
("id","resident","area","date","plastic","paper","glass","metal","organic","total","re
cycle%")
        self.tree = ttk.Treeview(tbl_card, columns=cols, show="headings",
                                 style="Custom.Treeview", selectmode="browse")

        widths = [70,170,160,100,80,80,75,75,90,85,95]
        heads  =
["ID","Resident","Zone","Date","Plastic","Paper","Glass","Metal","Organic","Total","♻
Rate"]
        for col, head, w in zip(cols, heads, widths):
            self.tree.heading(col, text=head)
            self.tree.column(col, width=w, anchor="center" if w < 100 else "w",
stretch=False)

        sb2 = ttk.Scrollbar(tbl_card, orient="vertical", command=self.tree.yview,
                            style="Custom.Vertical.TScrollbar")
        self.tree.configure(yscroll=sb2.set)
        self.tree.pack(side="left", fill="both", expand=True, padx=2, pady=2)
        sb2.pack(side="right", fill="y", pady=2)

        self._fill_table(self.data_manager.get_all_records())

    def _search_btn(self, text, cmd, bg, fg):
        return tk.Button(self.main, text=text, bg=bg, fg=fg,
                         activebackground=ACCENT, activeforeground=BG,
                         font=("Segoe UI", 11), bd=0, padx=14, pady=6,
                         relief="flat", cursor="hand2", command=cmd)
```

```
    def _fill_table(self, records):
        for item in self.tree.get_children():
            self.tree.delete(item)
        for r in records:
            w = r.get("waste", {})
            self.tree.insert("", "end", values=(
                r["collection_id"], r["resident"], r["area"], r["date"],
                f"{w.get('plastic',0):.1f}", f"{w.get('paper',0):.1f}",
                f"{w.get('glass',0):.1f}", f"{w.get('metal',0):.1f}",
                f"{w.get('organic',0):.1f}", f"{r['total_kg']:.1f}",
                f"{r['recyclable_pct']:.1f}%"
            ))

    def _do_search(self):
        q = self._search_var.get().strip()
        self._fill_table(self.data_manager.search_records(q))

    def _clear_search(self):
        self._search_var.set("")
        self._fill_table(self.data_manager.get_all_records())

    # ═══════════════════════════════════════════════════════════════
    # ADD COLLECTION FORM
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    def show_add_form(self):
        self.clear_main()
```

```
        self._page_header("Add New Collection", "Log a waste pickup — data is stored
across all DS structures")


        # Scroll wrapper
        outer = tk.Frame(self.main, bg=BG)
        outer.pack(fill="both", expand=True, padx=30, pady=(0, 25))

        canvas = tk.Canvas(outer, bg=BG, bd=0, highlightthickness=0)
        canvas.pack(side="left", fill="both", expand=True)
        sc = ttk.Scrollbar(outer, orient="vertical", command=canvas.yview)
        sc.pack(side="right", fill="y")
        canvas.configure(yscrollcommand=sc.set)

        inner = tk.Frame(canvas, bg=BG)
        win_id = canvas.create_window((0, 0), window=inner, anchor="nw")

        def _on_resize(e):
            canvas.itemconfig(win_id, width=e.width)
        canvas.bind("<Configure>", _on_resize)
        inner.bind("<Configure>", lambda e:
canvas.configure(scrollregion=canvas.bbox("all")))

        # ── Two-column form ──────────────────────────────────────────

        top_row = tk.Frame(inner, bg=BG)
        top_row.pack(fill="x", pady=8)

        left_card  = tk.Frame(top_row, bg=CARD_BG, padx=22, pady=18)
        right_card = tk.Frame(top_row, bg=CARD_BG, padx=22, pady=18)
```

```
        left_card.pack(side="left", fill="both", expand=True, padx=(0, 8))
        right_card.pack(side="left", fill="both", expand=True, padx=(8, 0))

        # — Left: Identity Fields —
        self._section_hdr(left_card, "Collection Identity")
        records = self.data_manager.get_all_records()
        next_id = f"WC{len(records) + 1:03d}"
        self.e_id  = self._form_field(left_card, "Collection ID",        next_id)
        self.e_res = self._form_field(left_card, "Resident / Collector Name", "")
        self.e_area = self._form_optionmenu(left_card, "Municipal Zone",
            ["Zone A (Block 1)", "Zone B (Block 2)", "Zone C (Block 3)",
             "Zone D (Market)", "Zone E (Industrial)"])
        self.e_date = self._form_field(left_card, "Collection Date (YYYY-MM-DD)",
                                       str(datetime.date.today()))
        self.e_notes = self._form_field(left_card, "Notes / Remarks (optional)", "")

        # — Right: Waste Breakdown —
        self._section_hdr(right_card, "Waste Breakdown (kg)")
        tk.Label(right_card, text="Enter the weight collected per category below:",
                 font=("Segoe UI", 10), bg=CARD_BG, fg=TEXT_SEC).pack(anchor="w",
pady=(0, 10))

        self.waste_entries = {}
        for cat in CATS:
            row = tk.Frame(right_card, bg=CARD_BG)
            row.pack(fill="x", pady=5)

            clr_dot = tk.Frame(row, bg=CAT_CLR[cat], width=10, height=10)
            clr_dot.pack(side="left", padx=(0, 8), pady=4)
```

```
        tk.Label(row, text=cat.capitalize(), font=("Segoe UI", 12),
                 bg=CARD_BG, fg=TEXT_PRI, width=9, anchor="w").pack(side="left")

        ent = tk.Entry(row, bg=CARD_BG2, fg=TEXT_PRI, insertbackground=ACCENT,
                       relief="flat", font=("Segoe UI", 12), width=10,
                       highlightthickness=1, highlightbackground=BORDER,
                       highlightcolor=ACCENT)
        ent.insert(0, "0.0")
        ent.pack(side="left", padx=8, ipady=5)
        tk.Label(row, text="kg", font=("Segoe UI", 10),
                 bg=CARD_BG, fg=TEXT_SEC).pack(side="left")
        self.waste_entries[cat] = ent

    # — Live Total Preview —
    self.total_var = tk.StringVar(value="Total: 0.0 kg")
    total_lbl = tk.Label(right_card, textvariable=self.total_var,
                         font=("Segoe UI", 14, "bold"), bg=CARD_BG, fg=ACCENT)
    total_lbl.pack(anchor="e", pady=(14, 0))

    def _update_total(*_):
        total = 0.0
        for cat, ent in self.waste_entries.items():
            try:
                total += float(ent.get() or 0)
            except ValueError:
                pass
        self.total_var.set(f"Total: {total:.2f} kg")
```

```
        for ent in self.waste_entries.values():
            ent.bind("<KeyRelease>", _update_total)

        # — Submit Button —
        submit_frame = tk.Frame(inner, bg=BG)
        submit_frame.pack(fill="x", pady=12)
        tk.Button(
            submit_frame,
            text="  💾   Save Collection Record",
            bg=ACCENT, fg=BG,
            activebackground="#00b390", activeforeground=BG,
            font=("Segoe UI", 13, "bold"),
            bd=0, padx=28, pady=12, relief="flat", cursor="hand2",
            command=self._submit_record
        ).pack(side="left")
        tk.Button(
            submit_frame, text="  🗑️   Clear Form",
            bg=CARD_BG, fg=TEXT_SEC, activebackground=BORDER,
            activeforeground=TEXT_PRI, font=("Segoe UI", 12),
            bd=0, padx=20, pady=12, relief="flat", cursor="hand2",
            command=self.show_add_form
        ).pack(side="left", padx=12)

    def _section_hdr(self, parent, text):
        tk.Label(parent, text=text, font=("Segoe UI", 13, "bold"),
                 bg=CARD_BG, fg=ACCENT).pack(anchor="w", pady=(0, 12))
        tk.Frame(parent, bg=BORDER, height=1).pack(fill="x", pady=(0, 12))
```

```
    def _form_field(self, parent, label, default):
        tk.Label(parent, text=label, font=("Segoe UI", 10),
                 bg=CARD_BG, fg=TEXT_SEC).pack(anchor="w", pady=(8, 2))
        ent = tk.Entry(parent, bg=CARD_BG2, fg=TEXT_PRI, insertbackground=ACCENT,
                       relief="flat", font=("Segoe UI", 12),
                       highlightthickness=1, highlightbackground=BORDER,
highlightcolor=ACCENT)
        ent.insert(0, default)
        ent.pack(fill="x", ipady=7, pady=(0, 4))
        return ent

    def _form_optionmenu(self, parent, label, options):
        tk.Label(parent, text=label, font=("Segoe UI", 10),
                 bg=CARD_BG, fg=TEXT_SEC).pack(anchor="w", pady=(8, 2))
        var = tk.StringVar(value=options[0])
        om  = tk.OptionMenu(parent, var, *options)
        om.configure(bg=CARD_BG2, fg=TEXT_PRI, activebackground=BORDER,
                     activeforeground=ACCENT, relief="flat",
                     font=("Segoe UI", 12), bd=0, highlightthickness=0)
        om["menu"].configure(bg=CARD_BG, fg=TEXT_PRI, activebackground="#1e3a5f",
                             activeforeground=ACCENT, font=("Segoe UI", 11))
        om.pack(fill="x", pady=(0, 4), ipady=4)
        return var

    def _submit_record(self):
        cid      = self.e_id.get().strip()
        resident = self.e_res.get().strip()
        area     = self.e_area.get()
```

```
        date     = self.e_date.get().strip()


        if not cid:
            messagebox.showerror("Validation", "Collection ID is required!")
            return
        if not resident:
            messagebox.showerror("Validation", "Resident / Collector name is
required!")
            return


        waste_dict = {}
        for cat, ent in self.waste_entries.items():
            try:
                waste_dict[cat] = max(0.0, float(ent.get()))
            except ValueError:
                waste_dict[cat] = 0.0


        if sum(waste_dict.values()) == 0:
            messagebox.showwarning("No Waste Data", "Please enter at least one waste
category amount.")
            return


        record = {"collection_id": cid, "resident": resident,
                  "area": area, "date": date, "waste": waste_dict}
        self.data_manager.add_record(record, record_undo=True, save_to_file=True)


        messagebox.showinfo("Success ✅",
                            f"Record '{cid}' for {resident} saved!\n"
                            f"Total: {sum(waste_dict.values()):.2f} kg collected.")
```

```
        self.show_records()


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    # SMART SUGGESTIONS

    #
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    def show_suggestions(self):

        self.clear_main()

        self._page_header("Smart Suggestions", "AI-driven eco tips and municipal
recommendations based on your data")


        records = self.data_manager.get_all_records()

        cat_totals = {c: 0.0 for c in CATS}

        for r in records:

            for cat in CATS:

                cat_totals[cat] += r.get("waste", {}).get(cat, 0)


        total_waste  = sum(cat_totals.values()) or 1

        plastic_pct  = cat_totals["plastic"]  / total_waste * 100

        organic_pct  = cat_totals["organic"]  / total_waste * 100

        paper_pct    = cat_totals["paper"]    / total_waste * 100

        metal_pct    = cat_totals["metal"]    / total_waste * 100

        recyclable   = (cat_totals["plastic"] + cat_totals["paper"] +

                        cat_totals["glass"]   + cat_totals["metal"]) / total_waste *
100

        avg_pickup   = (sum(r["total_kg"] for r in records) / len(records)) if records
else 0
```

```
        # Build suggestion cards
        suggestions = []

        if plastic_pct > 30:
            suggestions.append(("🧴", "High Plastic Volume Detected",
                f"{plastic_pct:.1f}% of total waste is plastic. Launch a plastic-free drive and "
                "partner with local vendors to reduce single-use packaging.",
                "#ef4444", "Critical"))

        if organic_pct > 40:
            suggestions.append(("🌱", "Set Up Composting Units",
                f"{organic_pct:.1f}% of waste is organic. Municipal compost bins or biogas plants "
                "could convert this to fertilizer or clean energy, reducing landfill load by ~35%.",
                "#a855f7", "High"))

        if recyclable < 50:
            suggestions.append(("♻️", "Boost Recycling Awareness",
                f"Only {recyclable:.1f}% of waste is recyclable material. Run educational campaigns "
                "on proper waste segregation (colour-coded bins: Blue=Dry, Green=Wet).",
                "#f59e0b", "Medium"))

        if avg_pickup > 20:
            suggestions.append(("🚛", "Optimize Pickup Frequency",
                f"Average pickup load is {avg_pickup:.1f} kg — exceeding recommended 15 kg/household. "
```

```
                "Increase collection rounds or deploy larger trucks in high-density
zones.",
                "#3b82f6", "High"))

        if paper_pct > 20:
            suggestions.append(("📄", "Paper Recycling Program",
                f"{paper_pct:.1f}% paper waste detected. Partner with recycling mills
_ "
                "every tonne of recycled paper saves 17 trees and 26,000 litres of
water.",
                ACCENT, "Medium"))

        if metal_pct > 15:
            suggestions.append(("🔩", "Metal Scrap Collection Drive",
                f"{metal_pct:.1f}% is metal waste. Organise quarterly scrap drives. "
                "Recycling metal saves up to 95% energy compared to producing from
ore.",
                ACCENT3, "Medium"))

        # General suggestions always shown
        suggestions.append(("📅", "Schedule Regular Audits",
            "Monthly waste data audits help identify area-wise trends early. "
            "Use the Dashboard Charts to compare zone performance and take proactive
action.",
            "#64748b", "General"))

        suggestions.append(("📲", "Digital Reporting & Resident Awareness",
            "Send monthly recycling reports to residents. Communities with regular
feedback "
            "reduce waste generation by an average of 22% within 6 months.",
            "#64748b", "General"))
```

```
        # Render
        scroll_outer = tk.Frame(self.main, bg=BG)
        scroll_outer.pack(fill="both", expand=True, padx=30, pady=(0, 25))

        canvas = tk.Canvas(scroll_outer, bg=BG, bd=0, highlightthickness=0)
        canvas.pack(side="left", fill="both", expand=True)
        sc = ttk.Scrollbar(scroll_outer, orient="vertical", command=canvas.yview)
        sc.pack(side="right", fill="y")
        canvas.configure(yscrollcommand=sc.set)
        sframe = tk.Frame(canvas, bg=BG)
        win = canvas.create_window((0, 0), window=sframe, anchor="nw")
        canvas.bind("<Configure>", lambda e: canvas.itemconfig(win, width=e.width))
        sframe.bind("<Configure>", lambda e:
canvas.configure(scrollregion=canvas.bbox("all")))

        PRIORITY_CLR = {"Critical": "#ef4444", "High": "#f59e0b",
                        "Medium": "#3b82f6", "General": "#64748b"}

        for icon, title, desc, accent_col, priority in suggestions:
            card = tk.Frame(sframe, bg=CARD_BG, padx=0, pady=0)
            card.pack(fill="x", pady=7)

            # Color stripe on left
            stripe = tk.Frame(card, bg=accent_col, width=5)
            stripe.pack(side="left", fill="y")

            body = tk.Frame(card, bg=CARD_BG, padx=18, pady=16)
            body.pack(side="left", fill="both", expand=True)
```

```
            title_row = tk.Frame(body, bg=CARD_BG)
            title_row.pack(fill="x")

            tk.Label(title_row, text=f"{icon}  {title}", font=("Segoe UI", 13,
"bold"),
                     bg=CARD_BG, fg=TEXT_PRI).pack(side="left")

            pri_color = PRIORITY_CLR.get(priority, "#64748b")
            badge = tk.Label(title_row, text=f"  {priority}  ",
                             font=("Segoe UI", 9, "bold"),
                             bg=pri_color, fg="white", padx=6, pady=2)
            badge.pack(side="right", padx=5)

            tk.Label(body, text=desc, font=("Segoe UI", 11), bg=CARD_BG,
                     fg=TEXT_SEC, wraplength=900, justify="left").pack(anchor="w",
pady=(6, 0))

    #
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    # DATA STRUCTURES INSPECTOR
    #
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    def show_ds_inspector(self):
        self.clear_main()
        self._page_header("Data Structures Inspector", "Live view into all active data
structures powering this app")

        tabview = ctk.CTkTabview(self.main, fg_color=CARD_BG,
```

```
                                segmented_button_fg_color="#0d1627",
                                segmented_button_selected_color=ACCENT,
                                segmented_button_selected_hover_color="#00b390",
                                segmented_button_unselected_color="#0d1627",
                                segmented_button_unselected_hover_color="#152040",
                                text_color=TEXT_PRI)
        tabview.pack(fill="both", expand=True, padx=30, pady=(0, 25))

        tabview.add("Doubly Linked List")
        tabview.add("Stack  (LIFO Undo)")
        tabview.add("Hash Table O(1)")
        tabview.add("Binary Search Tree")
        tabview.add("Priority Queue")

        def _textbox(parent, content):
            tb = ctk.CTkTextbox(parent, font=ctk.CTkFont(family="Consolas", size=11),
                                fg_color=CARD_BG2, text_color=TEXT_PRI,
                                scrollbar_button_color=BORDER)
            tb.pack(fill="both", expand=True, padx=8, pady=8)
            tb.insert("1.0", content)
            tb.configure(state="disabled")

        # ── Doubly Linked List ──────────────────────────────────────────────

        dll_content = (f"Head → {self.data_manager.dll.head.data['collection_id'] if self.data_manager.dll.head else 'NULL'}\n"
                       f"Tail → {self.data_manager.dll.tail.data['collection_id'] if self.data_manager.dll.tail else 'NULL'}\n"
                       f"Size  = {self.data_manager.dll.size}\n\n")
```

```
        curr = self.data_manager.dll.head
        idx  = 0
        while curr:
            prv = curr.prev.data['collection_id'] if curr.prev else "NULL"
            nxt = curr.next.data['collection_id'] if curr.next else "NULL"
            dll_content += (f"[{idx}]  ({prv}  ←  {curr.data['collection_id']}  →
{nxt})\n"
                            f"       Resident : {curr.data['resident']}\n"
                            f"       Area     : {curr.data['area']}\n"
                            f"       Total    : {curr.data['total_kg']} kg\n\n")
            curr = curr.next
            idx += 1
        _textbox(tabview.tab("Doubly Linked List"), dll_content)

        # ── Stack ──────────────────────────────────────────────────────────
        items = self.data_manager.undo_stack.to_list()
        stack_content = f"Stack Size: {self.data_manager.undo_stack.size}\n"
        stack_content += f"Top (Most Recent):
{self.data_manager.undo_stack.peek()}\n\n"
        if items:
            for i, item in enumerate(items):
                stack_content += f"[{i}]  Action: {item['action']}  →
{item['record']['collection_id']} ({item['record']['resident']})\n"
        else:
            stack_content += "Stack is currently empty — no undo history."
        _textbox(tabview.tab("Stack  (LIFO Undo)"), stack_content)

        # ── Hash Table ─────────────────────────────────────────────────────
```

```
        buckets = self.data_manager.resident_hash.get_buckets_info()
        ht_content = f"Total Keys Indexed: {self.data_manager.resident_hash.size}\n"
        ht_content += f"Capacity           :
{self.data_manager.resident_hash.capacity} buckets\n\n"
        for bucket_idx, keys in buckets:
            ht_content += f"Bucket [{bucket_idx:02d}]:  {'  →  '.join(keys) if keys
else '(empty)'}\n"
        _textbox(tabview.tab("Hash Table O(1)"), ht_content)

        # ── BST
──────────────────────────────────────────────────────────────────
        bst_records = self.data_manager.bst_index.in_order_traversal()
        bst_content = f"BST Node Count: {self.data_manager.bst_index.count}\n"
        bst_content += "In-Order Traversal (Sorted by Collection ID):\n\n"
        for key, record in bst_records:
            bst_content += (f"  Key: '{key}'  →  Resident: {record['resident']}  "
                            f"|  Weight: {record['total_kg']} kg  |  Area:
{record['area']}\n")
        _textbox(tabview.tab("Binary Search Tree"), bst_content)

        # ── Priority Queue
──────────────────────────────────────────────────────────
        heap_items = self.data_manager.pickup_pq.to_list()
        top = self.data_manager.pickup_pq.peek()
        pq_content = f"Heap Size    : {len(heap_items)}\n"
        pq_content += f"Top Dispatch: {top['collection_id'] if top else 'None'}\n"
        pq_content += f"Strategy    : Higher total weight = lower priority score =
dispatched first\n\n"
        for priority, data in heap_items:
            pq_content += (f"  Score [{priority:03d}]  →  {data['collection_id']}  |
 "
```

```
                        f"Area: {data['area']}  |  Weight: {data['total_kg']}
kg\n")
        _textbox(tabview.tab("Priority Queue"), pq_content)


    #
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    # UNDO
    #
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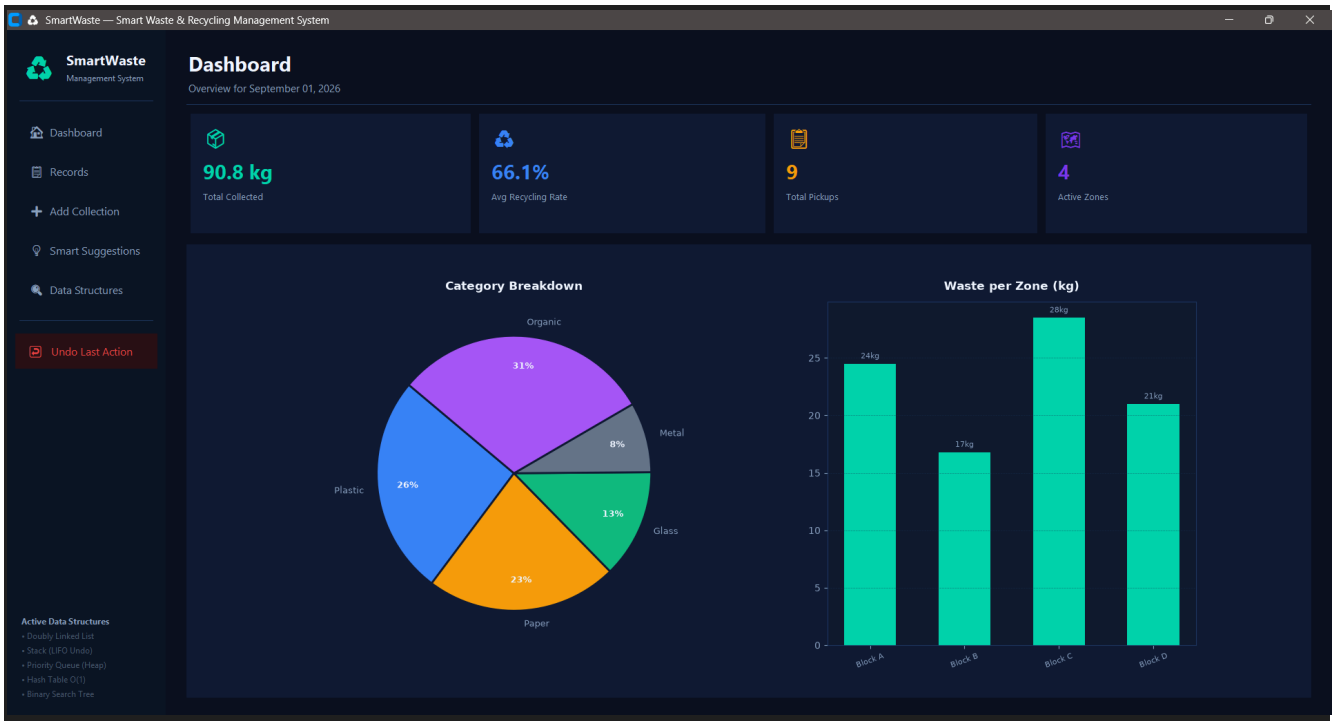
    def undo_action(self):
        success, msg = self.data_manager.undo_last_action()
        if success:
            messagebox.showinfo("Undo ↩", msg)
```


SmartWaste — Smart Waste & Recycling Management System
SmartWaste
Management System
Dashboard
Records
Add Collection
Smart Suggestions
Data Structures
Undo Last Action
Active Data Structures
• Doubly Linked List
• Stack (LIFO Undo)
• Priority Queue (Heap)
• Hash Table O(1)
• Binary Search Tree
Dashboard
Overview for September 01, 2026
90.8 kg
Total Collected
66.1%
Avg Recycling Rate
9
Total Pickups
4
Active Zones
Category Breakdown
Organic
31%
Metal
8%
Plastic
26%
Glass
13%
Paper
23%
Waste per Zone (kg)
28kg
24kg
21kg
17kg
25
20
15
10
5
0
Block A
Block B
Block C
Block D


SmartWaste — Smart Waste & Recycling Management System
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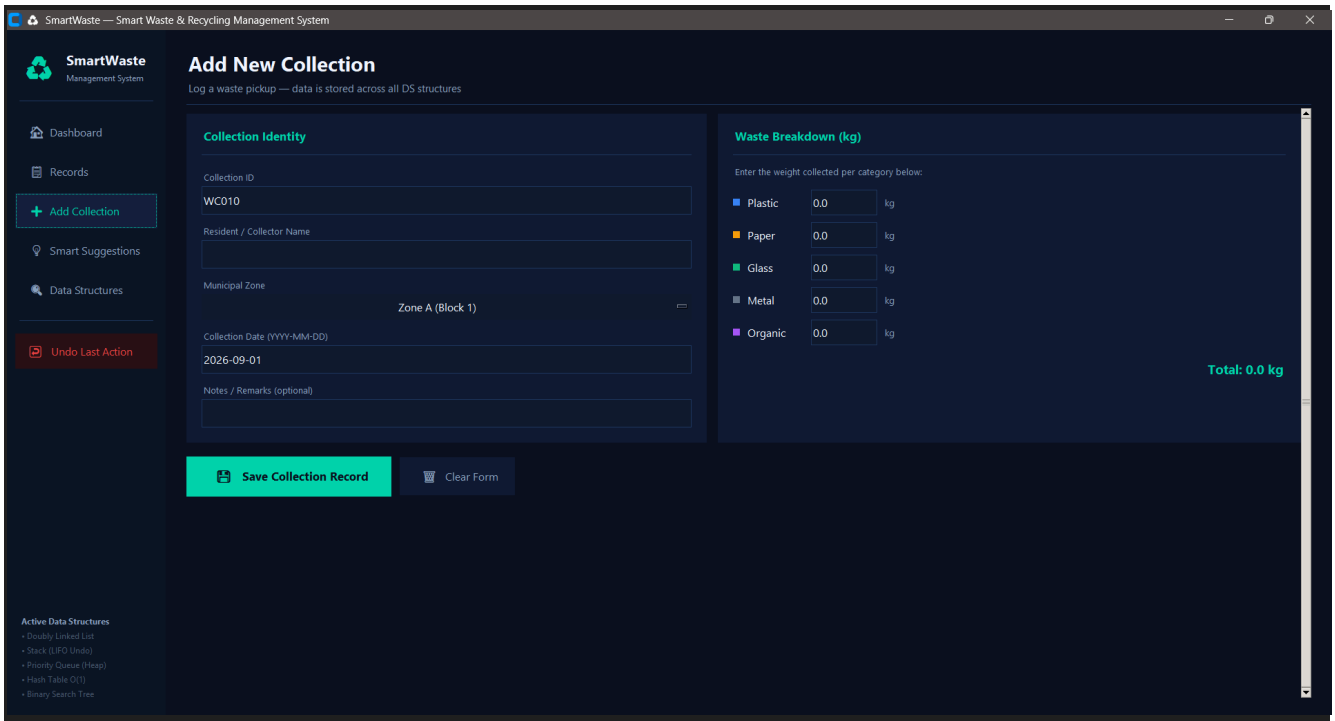
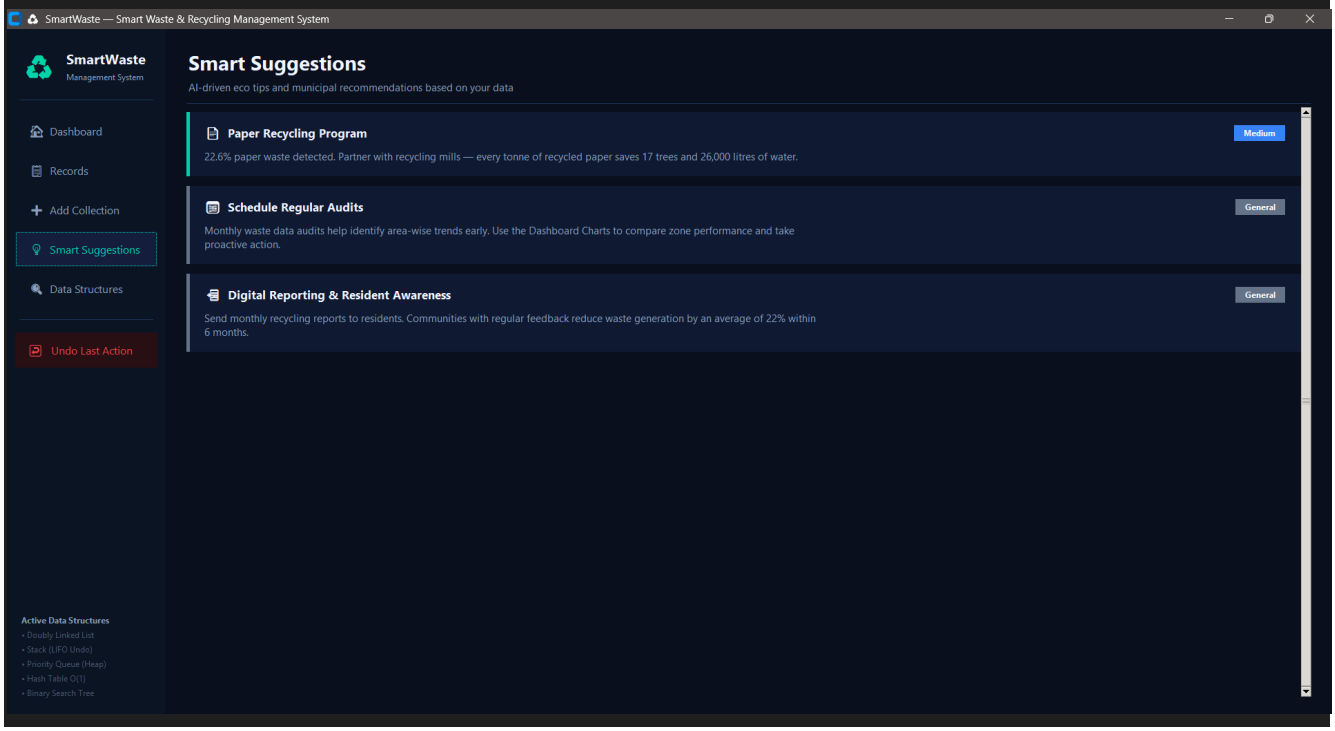
• Doubly Linked List
• Stack (LIFO Undo)
• Priority Queue (Heap)
• Hash Table O(1)
• Binary Search Tree
Records Manager
Browse, search and review all waste collection records
Search
Clear
ID Resident Zone Date Plastic Paper Glass Metal Organic Total Rate
WC001 Arun Kumar Block A 2026-08-25 2.5 1.5 1.0 0.5 3.0 8.5 64.7%
WC002 Priya Block B 2026-08-26 1.2 2.8 0.5 0.8 4.5 9.8 54.1%
WC003 Karthik Block C 2026-08-27 4.0 3.0 2.5 1.5 2.0 13.0 84.6%
WC004 Meena Block A 2026-08-28 0.8 1.0 0.0 0.2 5.0 7.0 28.6%
WC005 Rahul Block D 2026-08-29 3.5 2.0 1.8 1.2 3.5 12.0 70.8%
WC006 Divya Block B 2026-08-30 2.0 1.5 0.8 0.5 2.2 7.0 68.6%
WC007 Sanjay Block C 2026-08-31 5.0 4.0 3.0 2.0 1.5 15.5 90.3%
WC008 Anitha Block D 2026-09-01 1.5 2.2 1.0 0.3 4.0 9.0 55.6%
WC009 Vera Chen Block A 2026-09-02 3.0 2.5 1.0 0.5 2.0 9.0 77.8%


SmartWaste — Smart Waste & Recycling Management System
SmartWaste
Management System
Dashboard
Records
Add Collection
Smart Suggestions
Data Structures
Undo Last Action
Add New Collection
Log a waste pickup — data is stored across all DS structures
Collection Identity
Collection ID
WC010
Resident / Collector Name
Municipal Zone
Zone A (Block 1)
Collection Date (YYYY-MM-DD)
2026-09-01
Notes / Remarks (optional)
Waste Breakdown (kg)
Enter the weight collected per category below:
Plastic 0.0 kg
Paper 0.0 kg
Glass 0.0 kg
Metal 0.0 kg
Organic 0.0 kg
Total: 0.0 kg
Save Collection Record
Clear Form
Active Data Structures
• Doubly Linked List
• Stack (LIFO Undo)
• Priority Queue (Heap)
• Hash Table O(1)
• Binary Search Tree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AI-driven eco tips and municipal recommendations based on your data
Paper Recycling Program
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22.6% paper waste detected. Partner with recycling mills — every tonne of recycled paper saves 17 trees and 26,000 litres of water.
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Monthly waste data audits help identify area-wise trends early. Use the Dashboard Charts to compare zone performance and take proactive action.
Digital Reporting & Resident Awareness
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Send monthly recycling reports to residents. Communities with regular feedback reduce waste generation by an average of 22% within 6 months.
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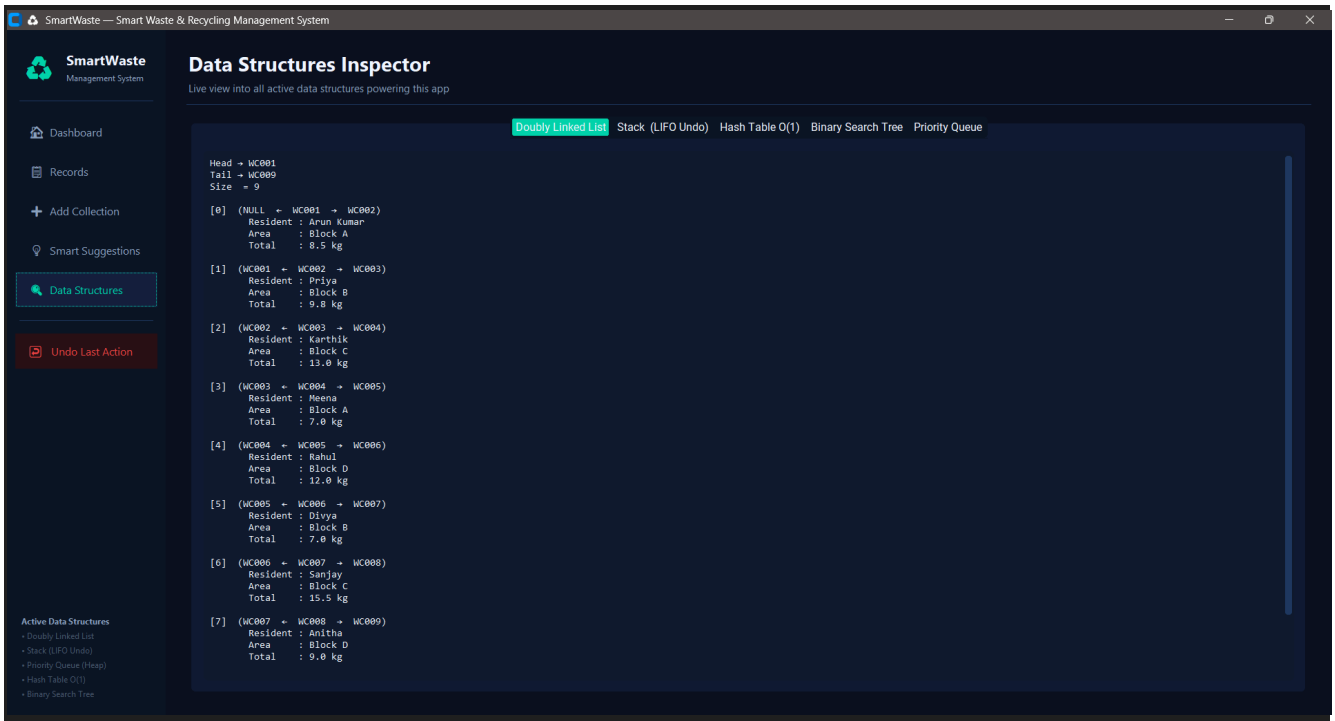

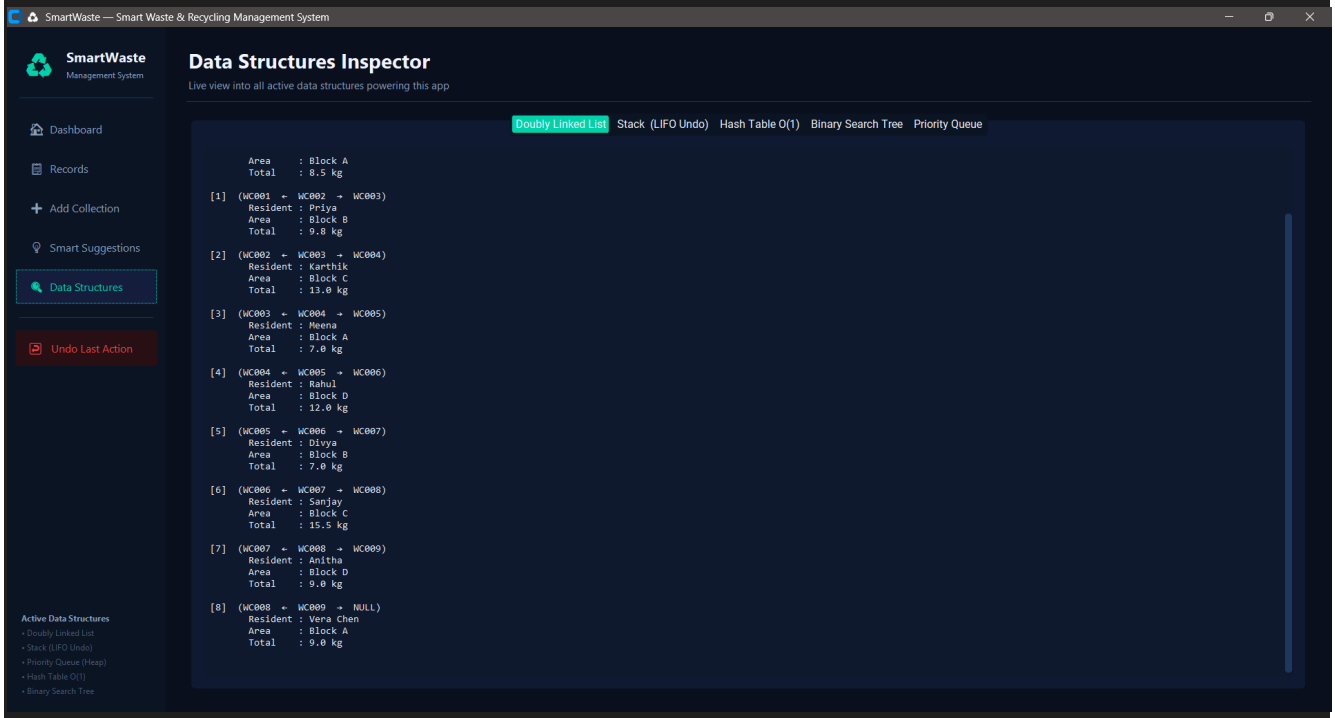


```
            self.show_dashboard()
        else:
            messagebox.showwarning("Undo ↩", msg)


# ——— Entry Point
```

```
if __name__ == "__main__":

    app = SmartWasteApp()

    app.mainloop()
```

IMPLEMENTATION SCREENSHOT: